

An Explainable Action-Token Generation Framework Based on Bladder Ultrasound Segmentation and Image Quality Assessment

Seong Jeong^{1,2,5}, Minsung Kim^{1,2,4}, Dongho Yee^{1,2,4}, Yechan Seo^{1,2,5}, Juahn Oh^{1,2,3}, Hyoun-Joong Kong^{1,2,5*}

¹ Department of Transdisciplinary Medicine, Seoul National University Hospital, Seoul, Republic of Korea

² Rosota Inc., Seoul, Republic of Korea

³ Eulji University College of Medicine, Daejeon, Republic of Korea

⁴ Department of Mechanical Engineering, Seoul National University, Seoul, Republic of Korea

⁵ Department of Medicine, Seoul National University, Seoul, Republic of Korea

*Corresponding Author

Abstract: During the morcellation phase of holmium laser enucleation of the prostate (HoLEP), a suprapubic ultrasound probe must be kept over the bladder lumen. Automating that view requires an explicit, auditable representation mapping image-derived states to actionable outputs; a segmentation mask alone is not such a representation. We present an explainable framework that converts bladder ultrasound segmentation and image-quality assessment into a safety-constrained, machine-readable action token. A U-Net segments the bladder lumen; a transparent image-quality function Q , computed from interpretable image and mask features, summarises the frame; and the system emits a structured token pairing a discrete action state, one of *move-left*, *move-right*, *hold*, or *check-filling*, with a continuous lateral displacement estimate $\hat{e}$, designed for deterministic decoding rather than natural-language recommendation. The vocabulary is bounded by what Q can distinguish: Q attains $[0.14, 0.78]$ of its nominal domain and, against frame-to-frame jitter, separates **47 distinguishable states**, the ceiling on safe action states. Two boundaries are derived, not tuned. A physical gate assigns *check-filling* because a non-anechoic lumen cannot be a filled bladder, and a statistical deadband assigns *hold* because an offset below 1.64σ of the centroid noise, $\sigma = 4.92$ px, cannot be signed. On 1,936 laterally translated held-out frames the action state is correct on 85.2% of frames at 0.10% direction error; the displacement stays continuous (slope 0.993), since quantising it into graded classes drops accuracy to 76.5% and 70.7%. The framework supplies an explainable action-token layer for HoLEP probe guidance; closed-loop validation on intraoperative imaging remains future work.

Keywords: bladder ultrasound segmentation, image quality assessment, explainable artificial intelligence, action token, robotic probe control, HoLEP morcellation

I. Introduction

During the morcellation phase of holmium laser enucleation of the prostate (HoLEP), a suprapubic ultrasound probe must be kept over the bladder lumen [1]. Automated analysis of that view needs an explicit representation of what the current image implies for an actionable state; a segmentation mask alone is not such a representation. Deep networks already segment the bladder accurately and cheaply [3, 4] and guidance systems can steer toward a standard view [5], but a mask is not something a downstream module can act on. An end-to-end learned mapping would produce motion directly yet hide *why* an action was issued and is hard to audit under segmentation uncertainty, whereas a raw geometric rule acts on the mask with no account of when the image is too degraded to act on at all. What is wanted instead is a structured layer that encodes the direction to move, withholds motion when the estimate is untrustworthy, and flags failures upstream of any probe motion.

We build that layer as an explainable **action-token generation framework** of three stages (Figure 1): a U-Net segments the bladder lumen; a transparent image-quality function Q , computed from interpretable image and mask features, summarises whether and how the frame falls short; and the state is emitted as a structured token pairing a discrete action state (*move-left*, *move-right*, *hold*, or *check-filling*) with a continuous lateral displacement estimate $\hat{e}$. The token is designed for deterministic decoding, with the identical human-readable wording retained only as an optional monitoring layer for oversight. The framework is confined to the lateral, image-preserving axis — the one axis presenting a measurable image-domain displacement that can be simulated from single-plane frames. This study validates the pipeline up to token generation; the token-to-command mapping is a design specification for future integration (Section IV), not a closed-loop result.

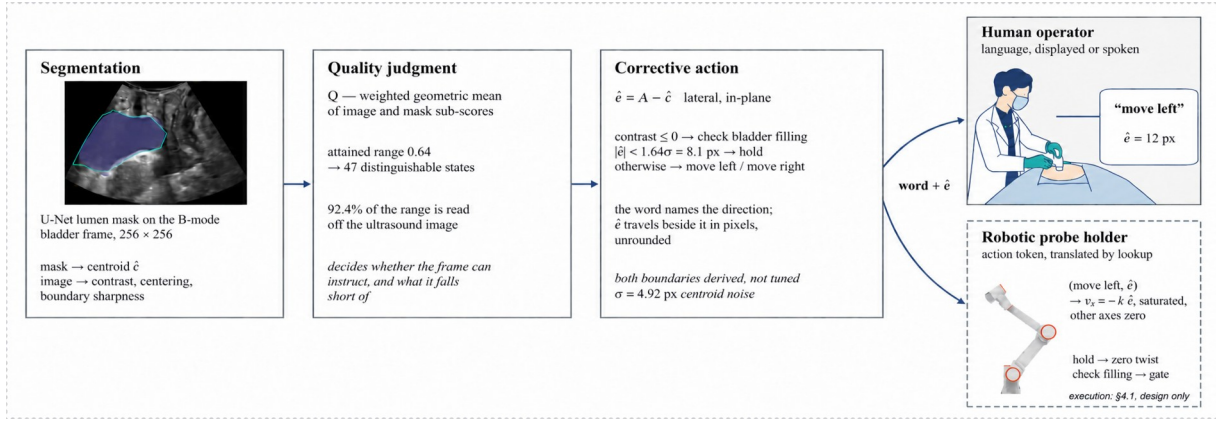


Figure 1. The action-token generation framework: an ultrasound frame is segmented (U-Net), summarised by the image-quality function Q , and emitted as a structured action token $(a, \hat{e}, Q)$ that a fixed decoder maps to a potential downstream system output. Stages 1–3 are validated; the dashed downstream execution (a potential future robotic-control integration) is not validated (Section IV).

II. Methods

1. Segmentation, Image State, and Quality Assessment

A U-Net [2] maps a 256×256 B-mode frame to a lumen probability map, thresholded at 0.5 and reduced to its largest hole-filled connected component. Frames come from PFUS1 [6], a public transperineal pelvic-floor dataset (101 women, midsagittal, rest and Valsalva, one scanner) whose eight annotated organs include the bladder [6, 7]; **it is not intraoperative HoLEP imaging**. The model was trained on a patient-disjoint split and used unchanged; all measurements use two held-out sets (1,661 and 1,292 frames) that share no patient. From the mask we compute an interpretable image state — centroid, area, boundary statistics, and lumen-to-surround contrast — every ratio taken over the insonified sector rather than the image rectangle.

The quality score is a weighted geometric mean of sub-scores $s_i \in [0, 1]$, every term and weight logged so a value is traceable to its components:

$$Q = \exp(\sum w_i \ln s_i / \sum w_i) \quad (1)$$

Dominated by its smallest term, Q cannot let one collapsed property hide inside an average; three image-reading terms (lumen contrast, centring, boundary sharpness) carry the top weight, and weights are tiered by rule, not fitted. As these are static examinations, the change in Q between adjacent frames estimates its own frame-to-frame jitter; the attained range divided by that jitter counts the states Q can distinguish — the ceiling on the number of reliable action states.

2. Action Token and Safety Boundaries

The token acts on the lateral, image-preserving axis v_x , the only plane-preserving axis presenting a measurable image-domain displacement. At frame t the framework emits a structured action token

$$\tau_t = (a, \hat{e}, Q_t) \quad (2)$$

where $a_t \in \{\text{move-left, move-right, hold, check-filling}\}$ is the discrete action state, $\hat{e}_t = A - \hat{c}_t$ is the continuous lateral displacement (beam axis A minus predicted lumen-centroid abscissa $\hat{c}$), and Q_t is the score of Equation (1). A fixed and auditable rule can decode the token for downstream use: the state selects an action mode, $\hat{e}$ supplies the continuous lateral scaling variable, and Q is available for conservative confidence-aware scaling, with no learned policy between the token and its downstream interpretation. A robotic lateral-velocity command is one potential future implementation of this decoding rule, but no hardware integration or closed-loop execution is evaluated in this study. Two derived, untuned boundaries fix the states (Table 1). The **physical gate** is exact: a urine-filled lumen is anechoic [8], so a non-positive lumen-to-surround contrast cannot be a filled bladder, and lateral motion cannot remedy insufficient filling. The **uncertainty deadband** follows from the centroid error $\delta = \hat{e} - e$ ($\sigma = 4.92$ px, bias $+0.11$ px): the estimated direction is wrong when δ opposes and exceeds e , a Gaussian-tail probability $\Phi(-|e|/\sigma)$; requiring the modelled error below 5% withholds motion under $1.64\sigma = 8.1$ px.

Table 1. Action-token specification: trigger, safety rationale, and deterministic control interpretation for each state (both boundaries derived, not tuned).

Action state	Trigger	Safety rationale	Control interpretation
<i>check-filling</i>	contrast ≤ 0	image failure, not a pose error	gate: no lateral motion
<i>move-left / move-right</i>	contrast $> 0, \hat{e} \geq 8.1$ px	direction reliable above noise	$v_x \propto \hat{e}$, other axes zero
<i>hold</i>	contrast $> 0, \hat{e} < 8.1$ px	direction within noise	deadband: zero lateral twist

3. Validation by Lateral Translation

In every recorded frame the lumen lies on one side of the beam axis, so e never changes sign and a constant output would score 99.3%. We therefore translate each frame laterally — the one probe motion single-plane frames reproduce faithfully — to

place e on prescribed targets, sampled densely near the axis crossing; fabricated margin is edge-replicated, never black. Segmentation is unaffected across the sweep (Dice 0.858), yielding 1,936 frames, 43% with $e < 0$.

III. Results

1. Segmentation and Quality Discrimination

Segmentation improves with bladder filling; in the working domain (annotated lumen $\geq 5.7\%$ of the sector, the state morcellation irrigation is meant to maintain) Dice reaches 0.882 (val) and 0.892 (test), and the lateral centroid error the token's

displacement rests on has a median of 3.85 and 3.65 px (Table 2). Below that domain segmentation degrades — exactly the population the *check-filling* state gates.

Q is defined on $[0, 1]$ but attains only $[0.14, 0.78]$ (width 0.64, central 80% 0.45); against a frame-to-frame jitter of 0.0095 this separates 67 states over the full range and **47 within the central 80%** — the ceiling on the number of reliable action states (Figure 2). Two image-reading terms carry 92.4% of the range, confirming that the discriminating information is image-derived rather than mask-derived.

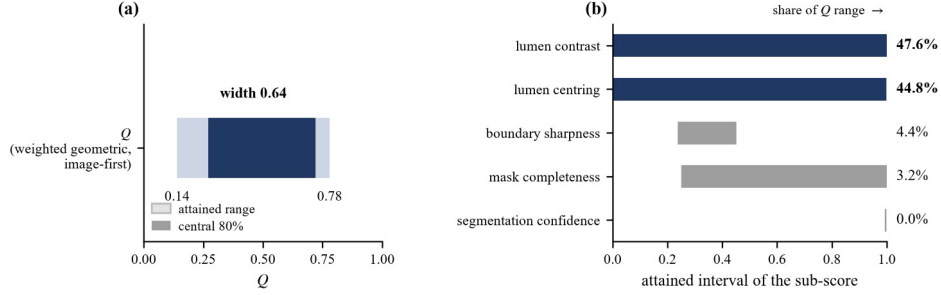


Figure 2. Quality-function discrimination: (a) the interval Q attains within its nominal $[0, 1]$ domain, with the central-80% operating band; (b) each active sub-score's attained interval and share of the range.

2. Action-Token Validation

On the 1,936-frame sweep the discrete action state is classified correctly on 85.2% of frames with a 0.10% direction error, and the continuous displacement is estimated at slope 0.993, $r = 0.970$, median error 3.15 px (Table 2, Figure 3). Wrong directions occur only near the axis — median required displacement 7.9 px against 15.8 px for correct ones — and their rate follows the Gaussian tail $\Phi(-|e|/\sigma)$, which is what licenses deriving the 8.1 px *hold* deadband from the model rather than reading it off the data (Figure 3).

Direction is thus robustly discretisable, whereas quantising the magnitude into graded action classes narrower than $2\sigma \approx 9.8$ px lowers classification accuracy to 76.5% (five states) and 70.7% (seven states) at unchanged direction error; the displacement is therefore kept as a continuous control variable rather than a class.

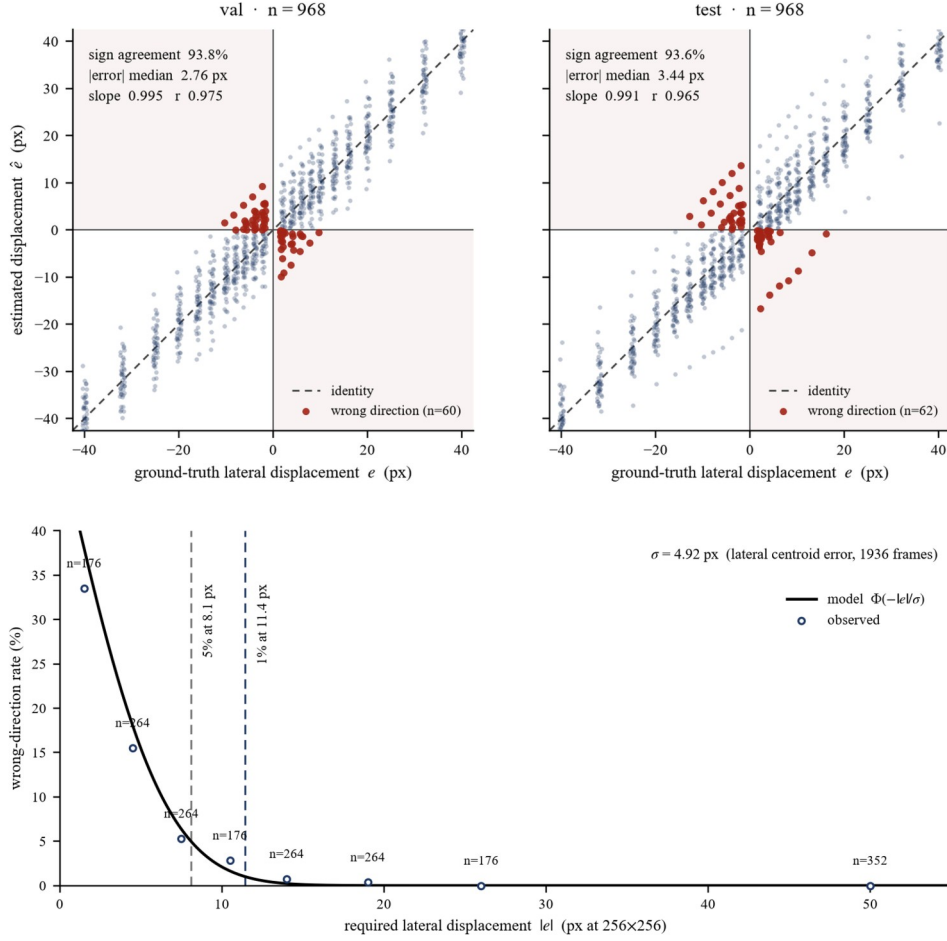


Figure 3. Action-token validation: (a) estimated versus ground-truth lateral displacement per held-out set (shaded quadrants are wrong-direction outcomes); (b) wrong-direction rate versus required displacement against the Gaussian-tail model $\Phi(-|e|/\sigma)$, with the modelled 5% and 1% boundaries marked.

Table 2. Main quantitative results: segmentation and action-token performance on the held-out and laterally translated frames.

Metric	Value
Segmentation Dice, all frames (val / test)	0.825 / 0.807
Segmentation Dice, working domain (val / test)	0.882 / 0.892
Lateral centroid error, working domain, median (val / test)	3.85 / 3.65 px
Attained Q range (width; central 80%)	0.64; 0.45
Distinguishable states (full; central 80%)	67; 47
Action-state accuracy (3 states); direction error	85.2%; 0.10%
Displacement estimate (slope; median error)	0.993; 3.15 px
Graded-magnitude accuracy (5; 7 states)	76.5%; 70.7%

IV. Discussion

An image-quality function intended to support structured downstream action handling should be judged by its attained range relative to its noise, because that ratio — not correlation with a segmentation metric — sets the number of safely distinguishable action states. A score can rank frames respectably while supporting fewer distinctions than a controller needs, a failure invisible to any accuracy figure; range and range-over-noise are cheap to compute and we suggest reporting them for any score intended to support downstream action handling.

The contribution is deliberately not to automate a judgment a trained observer makes by inspection — *move-left*, *move-right*, *hold*, and *check-filling* are visually obvious on screen, and we claim no diagnostic insight or improvement to human performance. It is to turn image-derived information into a structured, explainable, safety-constrained representation that can be interpreted deterministically by a downstream system, each token traceable to explicit features, segmentation geometry, a physical constraint, and measured uncertainty. The two non-motion states are the safety core: *hold* is an uncertainty-aware deadband that stops a controller from serving its own segmentation noise near the axis, and *check-filling* is a physics-based motion-inhibition gate marking that lateral repositioning is the wrong remedy — keeping an unreliable estimate, an upstream image failure, and a safety decision out of a black-box policy. The fixed decoding rule is an auditable integration specification for potential future systems, including robotic-control implementations, rather than a validated controller or hardware system.

Limitations. PFUS1 is transperineal pelvic-floor imaging, not intraoperative suprapubic HoLEP data; actual probe motion and closed-loop control were not validated, so the token-to-command mapping is a specification; and all target-domain constants — the

segmentation uncertainty σ and its derived boundaries — must be re-estimated before use in theatre.

V. Conclusion

An explainable, quality-aware action-token layer can bridge bladder ultrasound segmentation and structured downstream action handling. Its vocabulary is bounded by the measurable discrimination capacity of the image-quality function, and its *hold* and *check-filling* states supply explicit safety logic — an uncertainty-aware deadband and a physics-based gate — that a downstream system can interpret deterministically without a black-box policy. The study validates action-token generation and continuous displacement estimation on translated held-out frames; it does not validate closed-loop control or clinical deployment. Re-estimating the constants on intraoperative suprapubic HoLEP imaging and evaluating downstream integration remain future work.

References

1. T. Jang, H.-J. Kong, C. Baek, J. Kim, M. S. Choo, S.-J. Oh. Effect of Self-Training Using Virtual Reality Head-Mounted Display Simulator on the Acquisition of Holmium Laser Enucleation of the Prostate Surgical Skills. *International Neurourology Journal* 2024;28(2):138–146.
2. O. Ronneberger, P. Fischer, T. Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015)*, LNCS 9351, Springer, 2015, pp. 234–241.
3. M. Saini, Y. Jiang, T. Gangopadhyay, D. P. Rosen, A. Alizad, M. Fatemi. BWS-Net: An Optimal Deep Learning Architecture for the Anterior Bladder Wall Segmentation using Ultrasound Imaging. *IEEE Journal of Biomedical and Health Informatics* 2026. doi:10.1109/JBHL.2026.3675965.
4. Z. Song, M. Asiedu, S. Wang, Q. Li, A. Ozturk, V. Mittal, S. Schoen Jr., S. Ramaswamy, T. T. Pierce, A. E. Samir, Y. C. Eldar, A. Chandrakasan, V. Kumar. Memory-efficient low-compute segmentation algorithms for bladder-monitoring smart ultrasound devices. *Scientific Reports* 2023;13:16450.
5. H.-L. Hsu, M. Zahiri, G. Y. Li, R. Al Mukaddim, H. Lee, M. G. Wilson, J. Grube, S. Schmidt, G. Ghoshal, B. Raju. Active guidance in ultrasound bladder scanning using reinforcement learning. *Scientific Reports* 2026;16:5273.
6. D. Solís-Martín, J. A. Sainz, J. Galán-Páez, J. Borrego-Díaz, J. A. García-Mejido. PFUS1: Premier pelvic floor ultrasound segmentation dataset. A resource for advancing research. *Data in Brief* 2026;64:112346.
7. J. A. García-Mejido, D. Solís-Martín, M. Martín-Morán, C. Fernández-Conde, F. Fernández-Palacín, J. A. Sainz-Bueno. Applicability of deep learning to dynamically identify the different organs of the pelvic floor in the midsagittal plane. *International Urogynecology Journal* 2024;35(12):2285–2293.

8. Trinkler, Dietrich. Ultrasound of the Urinary Bladder. In: *EFSUMB Course Book, European Federation of Societies for Ultrasound in Medicine and Biology*, 2019.

[TO BE COMPLETED] Citations for HoLEP morcellation, ultrasound visual servoing and image-quality assessment are to be added before submission.